You have been supplied with the following source files:

StackData.txt
SecondStack.txt
HashData.txt

Open the evidence document, **evidence.doc**

Make sure that your name, centre number and candidate number will appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:
evidence_ followed by your centre number_candidate number, for example: **evidence_zz999_9999**

A class declaration can be used to declare a record. If the programming language used does not support arrays, a list can be used instead.

Three source files are used to answer **Questions 1** and **2**. The files are called **StackData.txt**, **SecondStack.txt** and **HashData.txt**

1 A program reads data from a text file and stores it in a stack. The stack `Stack` is stored as a 1D array of up to 20 elements. The pointer `TopOfStack` stores the index of the last element stored in the stack.

(a) `Stack` is a global array of strings with all elements initialised to `"-1"`

`TopOfStack` is a global variable initialised to `-1`

Write program code to declare and initialise `Stack` and `TopOfStack`

Save your program as **Question1_J25**. Copy and paste the program code into part **1(a)** in the evidence document.

[2]

(b) The function `Push()` takes a string parameter and attempts to store it on the stack.

The function returns `-1` if the stack is full.

The function returns `1` if the parameter is successfully pushed onto the stack.

Write program code for `Push()`

Save your program. Copy and paste the program code into part **1(b)** in the evidence document.

[4]